MATH6006: Statistical Methods for OR Modelling

Christine Currie

Mathematical Sciences

University of Southampton

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Practical Information

Lecturers

Statistics:

Professor Christine Currie, E-mail: christine.currie@soton.ac.uk

Stochastic Processes:

Dr Vesna Perisic, E-mail: V.Perisic@soton.ac.uk

Practical Information

Lecturers

Statistics:
Professor Christine Currie, E-mail: christine.currie@soton.ac.uk
Stochastic Processes:
Dr Vesna Perisic, E-mail: V.Perisic@soton.ac.uk

Lectures

Wednesday (L1) for Stochastic Processes
Friday (L2) for Statistics
Tuesday (PC) for both parts
Watch announcements on Blackboard module site.

Practical Information

Lecturers

Statistics:
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Aim

- To provide you with sufficient knowledge of the subject to enable you to carry out simple statistical procedures and understand statistical ideas and concepts

Syllabus

1. Organizing and displaying data; summary measures.
2. Probability distributions; discrete distributions, continuous distributions ; QQ-plot.
3. Central Limit Theorem and its applications; parameter estimation (method of moments); confidence intervals for mean and variance; hypothesis testing; comparison of two means; comparison of two variances; Chi-Square Tests.
4. Linear regression and basic ANOVA tests.
5. Practical statistics: the analysis of data and interpretation of the results.
6. Design of technical posters.

Learning Outcomes

On successful completion of the module, you should be able to:

- Define a number of common distributions, fit them to data and test the goodness of fit.
- State and apply the Central Limit Theorem.
- Derive confidence intervals for mean and variance and compare the means and variances using appropriate hypothesis tests.
- Conduct simple regression analysis and basic ANOVA tests and assess the applicability of the models used to the data.
- Interpret and present the results of the statistical techniques taught on the module
- Display technical material on a poster.

Assessment

Assessment (Statistical and Stochastic Parts Together)

70% Closed Book Examination (covering both statistics and stochastic processes)

30% Coursework (a poster and a short technical report).

Resources

References

Prem S.Mann: *Introductory Statistics (Sixth Edition)*

Anderson, Sweeney, Williams, Freeman, Shoesmith: *Statistics for Business and Economics*

Walpole and Mayers: *Probability and Statistics for Engineers and Scientists*

Hoel: *Introduction to Mathematical Statistics*

The Penguin Dictionary of Statistics by David Nelson: good glossary of statistical terms.

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Software

Examples will be coded in R: see `https://www.r-project.org/`. We assume students will use R or Python when creating the poster. Please confirm with the course lecturer if you wish to use a different package.

Statistics and Data

Statistics is typically split into two main areas.

- *Mathematical statistics* develops and proves statistical theorems, rules and laws.
- *Applied statistics* uses these to solve real-world problems.

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- *Mathematical statistics* develops and proves statistical theorems, rules and laws.
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We will deal mainly with applied statistics, which splits into two main areas:

- *Descriptive statistics*: methods for organizing, displaying and describing data using tables, graphs and summary measures (mean, mode, median, etc.).
- *Inferential statistics*: methods that use sample results to make decisions about a population.

Data: Elements, Variables and Observations

Most data has three components:

1. *Elements*: the entities on which data are collected.
2. *Variables*: a characteristic of interest for the elements. Variables can be qualitative or quantitative.
3. *Observations*: the result of a measurement obtained for a particular element

Measurements collected on each variable for every element in a study comprise a *data set*.

Qualitative versus Quantitative

- *Qualitative* (categorical) *data* include labels or names used to identify an attribute of each element.
- *Quantitative data* require numerical values indicating how much or how many. They can be
  - **discrete** (number of houses, cars, . . .), or
  - **continuous** (height, weight, . . .).

A *qualitative variable* is a variable with qualitative data; a *quantitative variable* is a variable with quantitative data.

Example: Exam Marks

Table: Example: Data for Two Students

	Exam			
Name	1	2	3	**Grade**
David	90	95	92	A+
Mary	85	90	91	A

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- The two elements are "David" and "Mary".
- The variables are "Exam 1", "Exam 2", "Exam 3" and "Grade".
- For the Exam variables the observations are numerical; for the Grade variable they are categorical.

Why does the Type of Data Matter?

An appropriate statistical analysis depends on whether the variable is quantitative or qualitative.

The statistical analysis for **qualitative** data is rather limited:

- determine frequencies and relative frequencies;
- represent these using a bar chart (or occasionally a pie chart).

There are many more alternatives for the statistical analysis of **quantitative** data.

Aside: Cross-Sectional and Time Series Data

- *Cross-sectional data* are collected on different elements at approximately the same point in time.
- *Time series data* are collected over several time periods on the same element(s).

Population, Sample, Census

- The group of all elements of interest in a particular study is the *population*.
- The smaller group of the population selected for the study is the *sample*.
- A *census* is the process of conducting a survey to collect data on the entire population.
- A *sample survey* collects data from a sample.

The methods of statistical inference use sample results to make estimates or test hypotheses about the characteristics of a population.

Descriptive Statistics: Graphical Data Presentations

Even a small set of data is often difficult to interpret straight from its values.

These so-called *raw data* are organised into tables and displayed using plots and other graphics.

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Data set

This example data set includes 50 individuals with one of the six most common last names in Ireland. The data file `names.txt` is on Blackboard.

We can draw a bar chart to show the frequencies of last names within the data set.

Example: Last Names — R Code

```r
# Read in data
names <- readLines("names.txt")
field_name <- "last_names"
names_df <- data.frame(name = names[-1], stringsAsFactors = FALSE)
colnames(names_df) <- field_name

# Find frequencies
frequency_df <- as.data.frame(table(names_df[[1]]))
colnames(frequency_df) <- c(field_name, "Frequency")

# Draw a bar plot with lime green coloured bars
ggplot(frequency_df, aes(x = last_names, y = Frequency)) +
  geom_bar(stat = "identity", fill = "limegreen") +
  labs(
    title = "Frequency of Last Names",
    x = names[1],
    y = "Frequency"
  )
```

Qualitative Data: Bar Chart

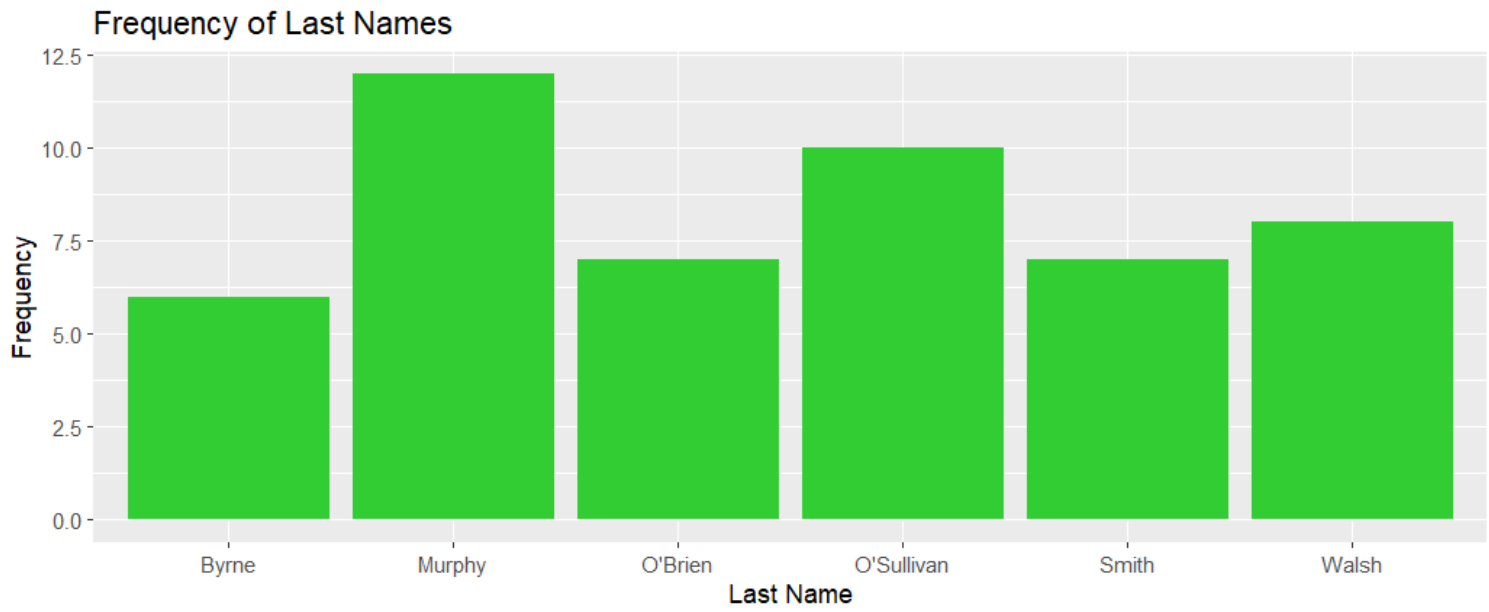


Figure: Bar Chart of Last Names

Summarising Qualitative Data

For a **qualitative** data set we can construct *frequency* and *relative frequency* distributions.

- A *frequency distribution* is a tabular summary showing the number (frequency) of items in each of several non-overlapping classes. Often displayed using a *bar chart*.
- A *relative frequency distribution* displays the frequency of each class divided by the number of elements in the whole data set. Often displayed using a *pie chart*.

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With n observations,

$$\text{Relative frequency of class} = \frac{\text{Frequency of class}}{n}.$$

Classes for Quantitative Data

With **quantitative** data we must be more careful in defining classes.

- A class is defined by its upper and lower limits: a data point belongs to a class if its value lies between those limits.
- There should be no overlap between classes.
- In the majority of examples, using the same width for each class works best.
- The number of classes depends on the number of data points and their range; typically between 5 and 20 classes are used.

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- In the majority of examples, using the same width for each class works best.
- The number of classes depends on the number of data points and their range; typically between 5 and 20 classes are used.

Choosing the number of classes requires judgment — a decision that best displays any natural grouping and variation in the data.

The relative frequency is defined as before and represents the proportion of observations belonging to a specific class.

Histograms

A *histogram* is a graphical representation of quantitative data showing the frequency or relative frequency of data values within classes. It is useful for understanding the shape or *skewness* of a numerical variable.

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How does a histogram differ from a bar chart?

- A histogram displays numerical data, whereas a bar chart is typically used for categorical data.
- The frequency of each class is shown by a rectangle whose base is determined by the class limits on the horizontal axis, and whose height is the corresponding frequency or relative frequency.
- Bars are adjacent.

Drawing a Histogram

$$\text{Relative Frequency} = \frac{\text{Number of observations within a group}}{\text{Number of observations in total}}$$

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$$\text{Height of bar} = \frac{\text{(Relative) frequency}}{\text{Width of interval}}$$

$$\text{Area of bar} = \text{Width of interal} \times \frac{\text{(Relative) frequency}}{\text{Width of interval}} = \text{(Relative) Frequency}$$

Drawing a Histogram

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$$\text{Area of bar} = \text{Width of interal} \times \frac{\text{(Relative) frequency}}{\text{Width of interval}} = \text{(Relative) Frequency}$$

- The sum of the bar areas is 1 for relative frequency or the number of datapoints for actual frequency.

Example: Computer Usage — Histograms

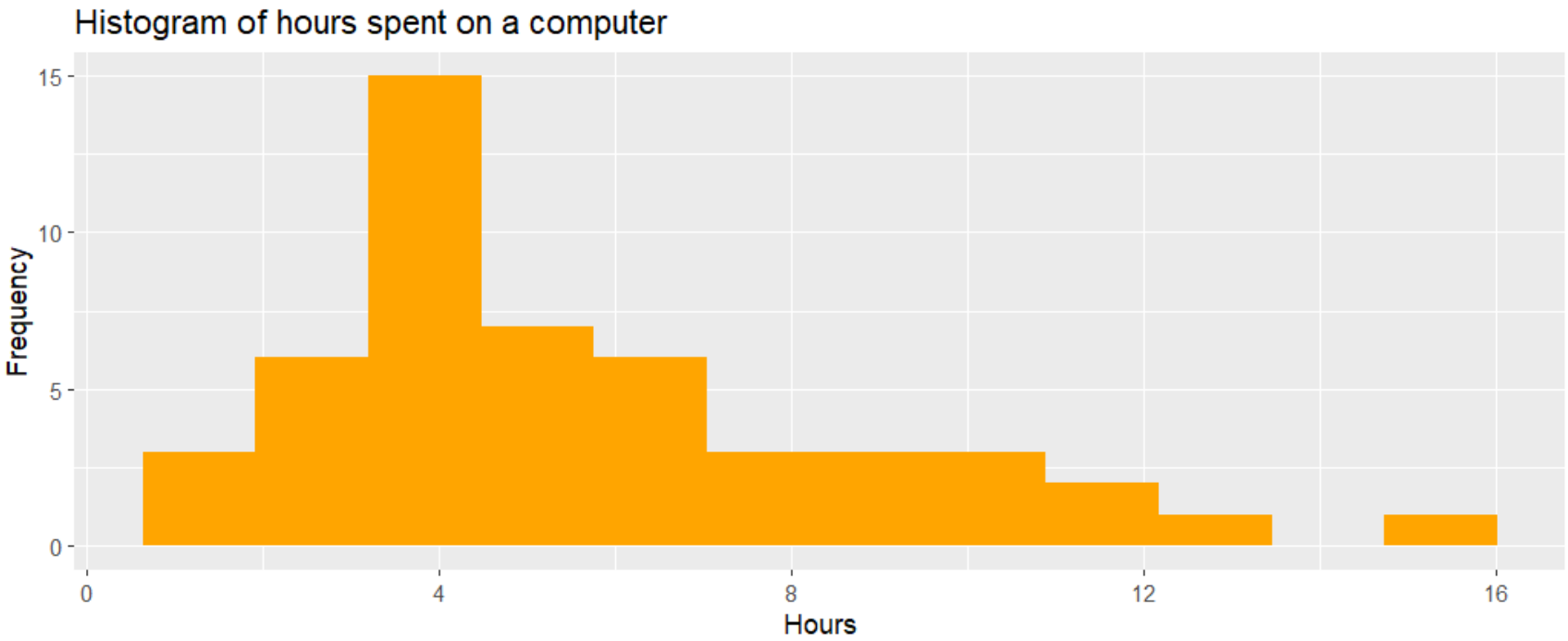
Data set

Data on personal computer usage by persons aged 12 and older during one week, for a sample of 50 persons, in hours. The data are in `computer.txt` on Blackboard.

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Example: Computer Usage — R Code

```r
# Read in computer data
computer <- readLines("computer.txt")
computer_field_name <- "Hours"
computer_df <- data.frame(computer = computer[-1])
colnames(computer_df) <- computer_field_name
computer_df$Hours <- as.double(computer_df$Hours)

# Draw histogram
ggplot(computer_df, aes(x = Hours)) +
  geom_histogram(
    bins = 12,
    fill = "orange"
  ) +
  labs(
    title = "Histogram of hours spent on a computer",
    x = "Hours",
    y = "Frequency"
  )
```

Cumulative Frequency Distribution

- The *cumulative frequency distribution* shows the number of data items with values less than or equal to the upper class limit of each class.
- The *cumulative relative frequency distribution* shows the proportion of data items with values less than or equal to the upper limit of each class.
- A graph of a cumulative distribution is called an *ogive*: data values on the horizontal axis, cumulative frequencies or cumulative relative frequencies on the vertical axis.

We return to this later in the chapter when we introduce the cumulative distribution function.

Empirical CDF — R Code

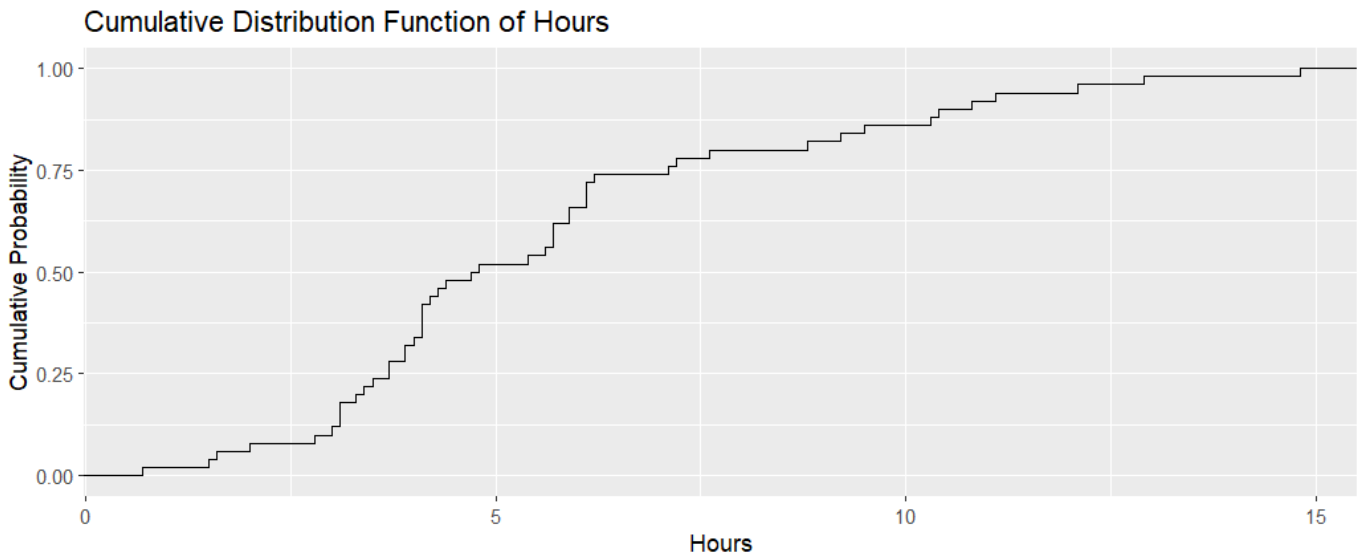
For example, we can plot the empirical cumulative distribution function (ECDF) for the computer example.

```
ggplot(computer_df, aes(x = Hours)) +
  stat_ecdf() +
  labs(
    title = "Cumulative Distribution Function of Hours",
    x = "Hours",
    y = "Cumulative Probability"
  )
```

Cumulative Distribution Function for the Computer Example

CDF: Cumulative Distribution Function

CDF shows the proportion of data items with values below the upper limit of each class

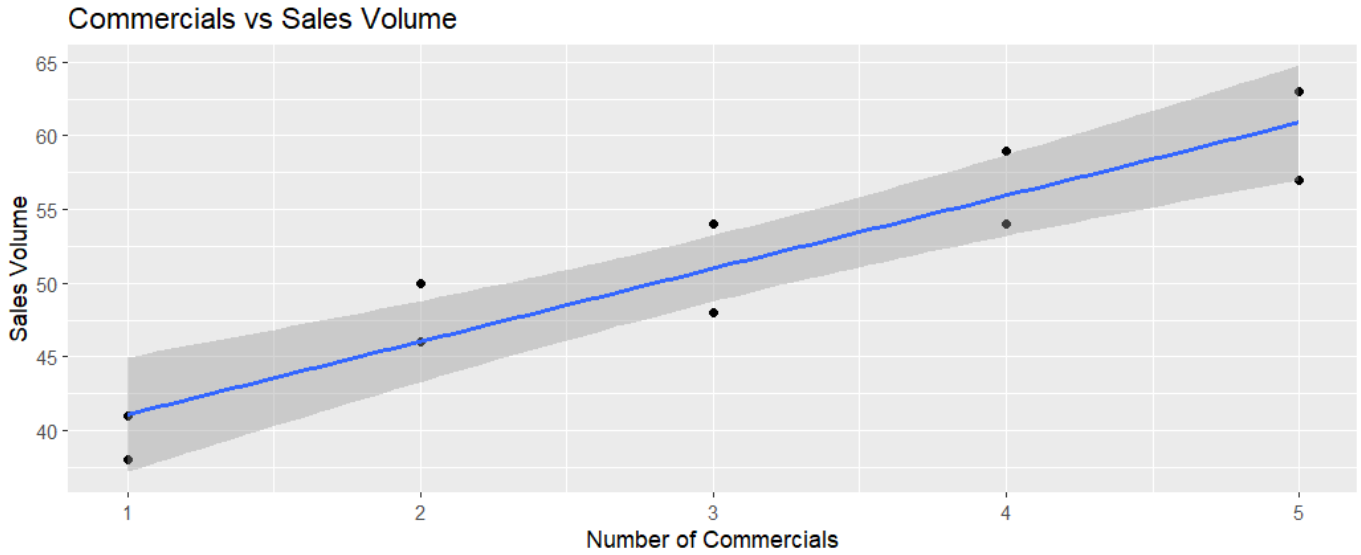


Scatter Plot and Trend Line

A *scatter plot* shows the relationship between two variables and a *trend line* is an approximation of the relationship

Example: Computer Store

The data have three columns: the week number, the number of commercials shown, and the volume of sales at the equipment store that week. Available on Blackboard as `store.txt`.



Computer Store — R Code

```
# Read in equipment store data
equipment_df <- read.table(
  "store.txt",
  header = TRUE,
  sep = "\t"
)

ggplot(equipment_df, aes(x = `No..of.Commercials`, y = `Sales.Volume`)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    title = "Commercials vs Sales Volume",
    x = "Number of Commercials",
    y = "Sales Volume"
  )
```

Numerical Descriptive Measures

Numerical descriptive measures are measures of **location**, **dispersion**, **shape** and **association**.

- Calculated for sample data, they are called *sample statistics*.
- Calculated for population data, they are called *population parameters*.
- A sample statistic is the *point estimator* of the corresponding population parameter.

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Notation convention

- X (capital) denotes a variable, whose observations can be made.
- x (small) denotes a generic value that the variable X may take.

For example, X could be the height of people, and $x = 185\,\mathrm{cm}$ is a value of X for a particular person.

Measures of Location: The Mean

The *mean* (average value) is the most frequently used measure of central location.

- The **sample mean** of $\mathrm{x} = (x_1, x_2, \ldots, x_n)$ is denoted $\bar{x}$.
- The **population mean** of $\mathrm{x} = (x_1, x_2, \ldots, x_N)$ is denoted μ.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \qquad \text{and} \qquad \mu = \frac{\sum_{i=1}^{N} x_i}{N},$$

where n is the sample size, N the population size, and x_i the value of the variable X on the i-th observation.

The expression for the mean includes all values of the data set and, as a result, it is **sensitive to outliers**.

Measures of Location: Median and Mode

Median

The value of the "middle" term when the data are arranged in increasing order. If n is odd it is the middle value; if n is even it is the average of the two middle values. The median is **not influenced by outliers**.

Measures of Location: Median and Mode

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Mode

The value in a data set that occurs with the highest frequency. It can be determined for both quantitative and qualitative data.
A data set can have no mode (all values equally frequent), one mode, or several modes. The modes of multimodal data are not very helpful.

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The value in a data set that occurs with the highest frequency. It can be determined for both quantitative and qualitative data.
A data set can have no mode (all values equally frequent), one mode, or several modes. The modes of multimodal data are not very helpful.

Deciding which measure to use (mean, median or mode) depends on your data set.

Example: Trades of Lloyds Bank Group Shares

The last 5 trades of Lloyds Bank Group shares on 22 September 2017 (data from the London Stock Exchange).

Time	Price	Volume	Trade value
17:11:37	66.98	93,163	62,400.53
17:11:29	67.03	80,204	53,760.78
17:11:29	67.03	13,178	8,833.22
17:03:06	67.15	70,800	47,542.20
17:02:36	67.00	9,064	6,072.64

Lloyds Example: Mean, Median and Mode

The **mean** price is

$$\bar{p} = \frac{66.98 + 67.03 + 67.03 + 67.15 + 67.00}{5} = 67.038 \approx 67.04 \text{ (2 d.p.)}.$$

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The **mean** price is

$$\bar{p} = \frac{66.98 + 67.03 + 67.03 + 67.15 + 67.00}{5} = 67.038 \approx 67.04 \text{ (2 d.p.)}.$$

Here $n = 5$, so the **median** is the third highest price. The ordered list is $(66.98,\ 67.00,\ 67.03,\ 67.03,\ 67.15)$, and the third value is 67.03.

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Here $n = 5$, so the **median** is the third highest price. The ordered list is $(66.98,\ 67.00,\ 67.03,\ 67.03,\ 67.15)$, and the third value is 67.03.

The price 67.03 appears twice, so this is the **mode**.

Measures of Variability

The most common measures of variability are the **range**, **interquartile range**, **variance** and **standard deviation**.

- The *range* is the difference between the largest and smallest values. It is based on only two observations and is highly influenced by outliers.
- The *interquartile range* is the range of the middle 50% of the data, and so does not depend on outliers.
- The *variance* takes into account all of the data. It is based on the *deviation about the mean*,

$$x_i - \bar{x}.$$

Variance and Standard Deviation

Population variance σ^2, for a population of size N with mean μ:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N-1}.$$

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$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}.$$

The statistic s^2 gives an unbiased estimate of the population variance. Note the divisor $(n-1)$, not n as one might expect.

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The variance is useful when comparing variability between two or more variables, samples or populations: the one with the largest variance shows the greatest variability.

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The variance is useful when comparing variability between two or more variables, samples or populations: the one with the largest variance shows the greatest variability.

The *standard deviation* is the positive square root of the variance. It is measured in the same units as the original data, and is consequently common in practice.

Coefficient of Variation

The *coefficient of variation* measures how large the standard deviation is relative to the mean:

$$\frac{\text{Standard deviation}}{\text{Mean}} \times 100\%.$$

It is a descriptive statistic that indicates the size of the standard deviation relative to the mean, and is independent of the units of the data.

Percentiles

A *percentile* provides information on how the ordered data are spread over the range of the data. The p-th percentile is a value such that at least p percent of the data are less than or equal to it, and at least $(100 - p)$ percent are greater than or equal to it.

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Calculating percentiles for a sample $x_1, x_2, \ldots, x_n$

1. Arrange the data in increasing order, $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$.
2. Calculate $i = \left(\frac{p}{100}\right) n$, where p is the percentile of interest and n the sample size.
3. If i is **not** an integer, round up; the resulting value of i gives the position of the p-th percentile. If i **is** an integer, the p-th percentile is the average of the values in positions i and $i + 1$.

Example: Waiting Time

Waiting List Data set

The waiting time data is a list of 24 values describing the time spent waiting for a service. They can be found in the file `WaitingTime.txt`.

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(a) Compute the 30th percentile. Here $n = 24$ and $p = 30$:

$$i = \left(\frac{p}{100}\right) \times n = \frac{30}{100} \times 24 = 7.2.$$

Hence the 30th percentile is

$$x_{(\lceil i \rceil)} = x_{(\lceil 7.2 \rceil)} = x_{(8)} = 30,$$

where $\lceil i \rceil$ is the smallest integer greater than or equal to i.

Example: Waiting Time — Quartiles

(b) Compute the 25th percentile. Here $n = 24$, $p = 25$:

$$i = \frac{25}{100} \times 24 = 6 \quad \text{(an integer)},$$

$$Q_1 = \frac{x_{(i)} + x_{(i+1)}}{2} = \frac{x_{(6)} + x_{(7)}}{2} = \frac{26 + 28}{2} = 27.$$

Example: Waiting Time — Quartiles

(b) Compute the 25th percentile. Here $n = 24$, $p = 25$:

$$i = \frac{25}{100} \times 24 = 6 \quad \text{(an integer)},$$

$$Q_1 = \frac{x_{(i)} + x_{(i+1)}}{2} = \frac{x_{(6)} + x_{(7)}}{2} = \frac{26 + 28}{2} = 27.$$

(c) Compute the 75th percentile. Here $n = 24$, $p = 75$:

$$i = \frac{75}{100} \times 24 = 18 \quad \text{(an integer)},$$

$$Q_3 = \frac{x_{(18)} + x_{(19)}}{2} = \frac{61 + 62}{2} = 61.5.$$

Quartiles

The *quartiles* Q_1, Q_2 and Q_3 divide an ordered data set into four equal parts.

- Q_1 is the 25th percentile: 25% of the data set is equal to or smaller than Q_1.
- Q_2 is equal to the median.
- Q_3 is the 75th percentile.
- The *interquartile range* is $\mathrm{IQR} = Q_3 - Q_1$.

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- Q_3 is the 75th percentile.
- The *interquartile range* is $\mathrm{IQR} = Q_3 - Q_1$.

Steps to calculate the quartiles

1. Arrange the observations in non-decreasing order and locate the median M.
2. Q_1 is the median of the observations whose positions in the ordered list lie to the left of M.
3. Q_3 is the median of the observations whose positions in the ordered list lie to the right of M.

The Five-Number Summary and Boxplots

The five-number summary of a set of observations consists of

$$\text{Minimum}, \quad Q_1, \quad M, \quad Q_3, \quad \text{Maximum}.$$

These are well represented in a *boxplot*:

- a central box spanning the quartiles Q_1 and Q_3;
- a line in the box marking the median M;
- lines extending from the box out to the lower and upper limits using the following rules:

$$\begin{aligned} \text{lower limit} &= \max\{\text{Minimum}, M - 1.5 \times \text{IQR}\} \\ \text{upper limit} &= \min\{\text{Maximum}, M + 1.5 \times \text{IQR}\} \end{aligned}$$

- all other observations beyond the limits are marked as whiskers; they are suspected outliers.

Boxplot: Waiting Time Example

We have already calculated

$$Q_1 = 27, \quad M = 45.5, \quad Q_3 = 61.5,$$

so

$$\mathrm{IQR} = Q_3 - Q_1 = 61.5 - 27 = 34.5.$$

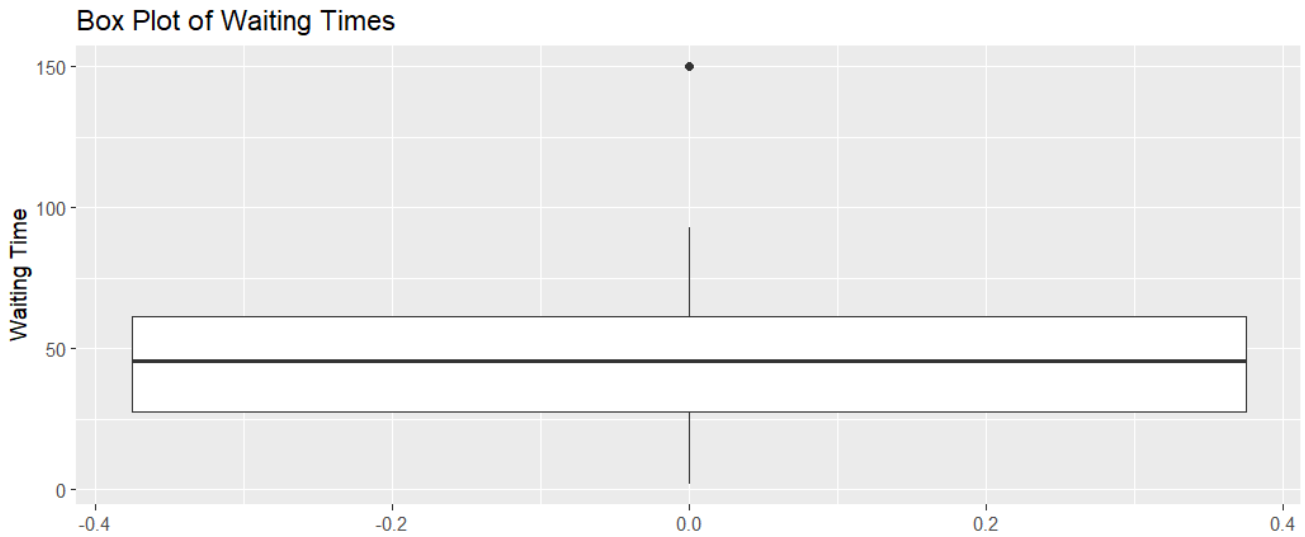
The lower limit is

$$\begin{aligned} L &= \max\{\text{Minimum}, M - 1.5 \times \text{IQR}\} \\ &= \max\{2, 45.5 - 1.5 \times 34.5 = -6.25\} = 2. \end{aligned}$$

and the upper limit is

$$\begin{aligned} U &= \min\{\text{Maximum}, M + 1.5 \times \text{IQR}\} \\ &= \min\{150, 45.5 + 1.5 \times 34.5 = 97.25\} = 97.25. \end{aligned}$$

Boxplot for the Waiting Time Data



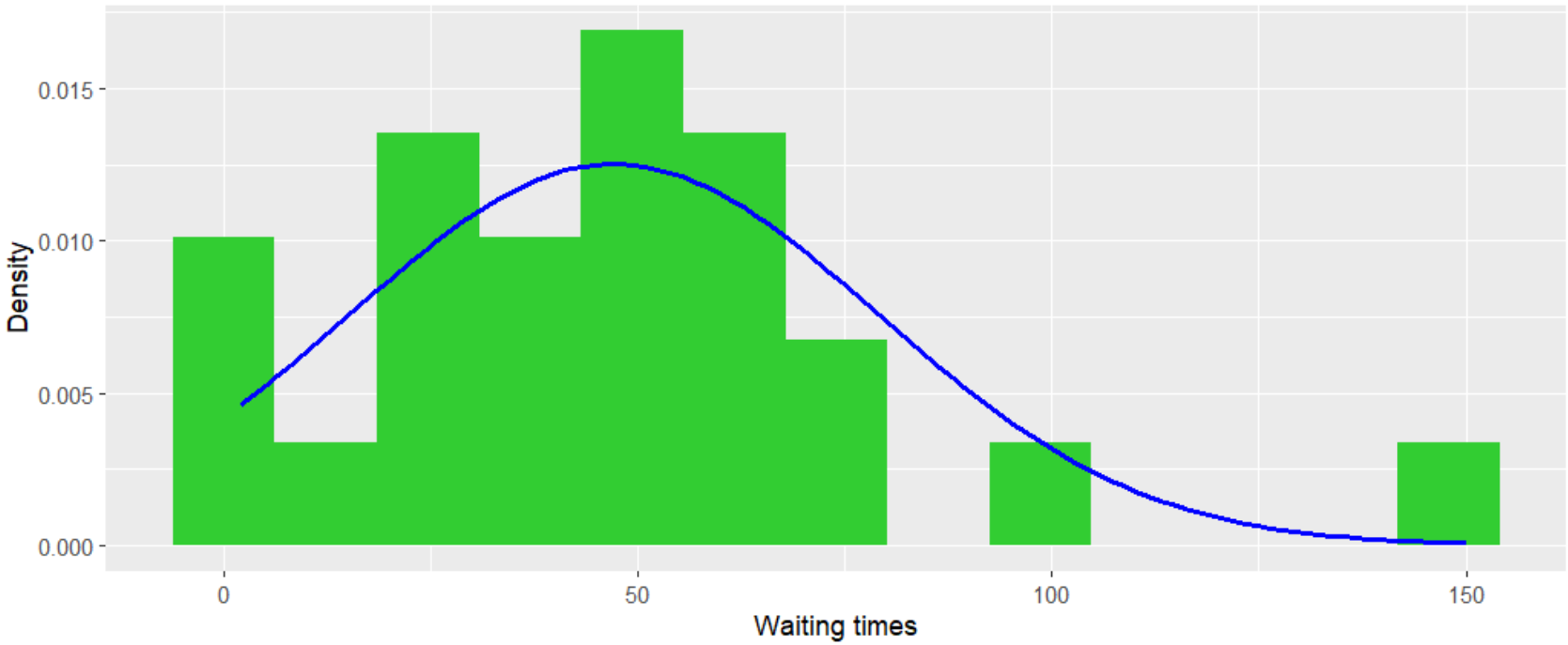
Boxplot for the waiting time data set

Probability Density Functions — R Code

A probability density function $f(x)$, or PDF (defined properly later), provides a more complete description of how the data are distributed. For example, we might fit a distribution to the waiting time data.

```
ggplot(waiting_df, aes(x = Waiting.Time)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 13,
    fill = "limegreen",
  ) +
  stat_function(
    fun = dnorm,
    args = list(
      mean = mean(waiting_df$Waiting.Time),
      sd = sd(waiting_df$Waiting.Time)
    ),
    colour = "blue",
    linewidth = 1
  ) +
  labs(x = "Waiting times", y = "Density")
```

Waiting Times with a Fitted Normal Distribution



Histogram of the waiting times data with a fitted normal distribution.

The fitted curve is completely determined by the sample mean ($\bar{x} = 47.04$) and the standard deviation ($s = 31.87$) of the data.

The Normal Distribution

The density function takes the sample mean as its true mean μ and the sample standard deviation as its true standard deviation σ:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

This is the **normal** (or Gaussian) distribution, often denoted $N(\mu, \sigma)$. Its location and shape are completely determined by (μ, σ), and the density is symmetric about μ.

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It is a density function because

1. $f(x) \geq 0$ for all $x \in \Re$;
2. the area underneath the curve is 1: $\int_{-\infty}^{+\infty} f(x)\, dx = 1$.

The 68–95–99.7 Rule

Although different (μ, σ) generate different curves, normal distributions share a common feature.

In the normal distribution with mean μ and standard deviation σ:

- approximately 68% of observations fall within σ of the mean μ;
- approximately 95% fall within 2σ of the mean μ;
- approximately 99.7% fall within 3σ of the mean μ.

The Standard Normal Distribution

Introduce a new variable Z such that

$$Z = \frac{X - \mu}{\sigma}, \qquad \text{equivalently} \qquad X = \sigma Z + \mu.$$

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It follows from $\int f(x)\,dx = 1$ over $(-\infty, \infty)$ that

$$\begin{aligned}
1 &= \int_{-\infty}^{+\infty} f(x)\,dx \\
&= \int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\
&= \int_{-\infty}^{+\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \sigma\, dz \\
&= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}\, dz.
\end{aligned}$$

The Standard Normal Probability Density Function

The function

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

is a density function corresponding to a normal distribution with mean 0 and standard deviation $\sigma = 1$.

Its distribution is denoted $N(0, 1)$ and is known as the **standard normal distribution**.

Cumulative Distribution Function (CDF)

Let X follow a distribution with pdf $f(x)$. The cumulative distribution at a given value x is

$$F(x) = \int_{-\infty}^{x} f(t)\, dt.$$

The CDF of the standard normal distribution is frequently used to estimate confidence intervals on estimates; we will see later why.

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Let $Z \sim N(0, 1)$ with density $\phi(z)$. Its cumulative distribution is

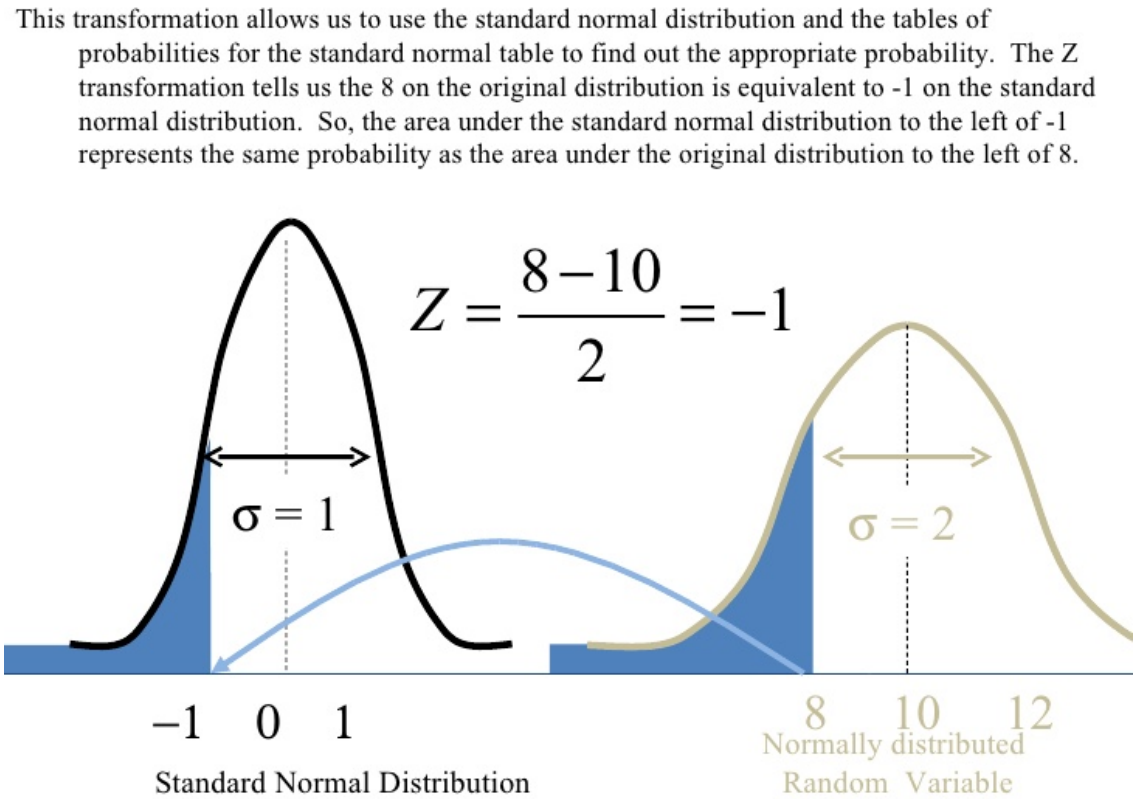
$$\Phi(z) = \int_{-\infty}^{z} \phi(t)\, dt.$$

It follows from the Z transformation that

$$F(x) = \Phi\left(\frac{x - \mu}{\sigma}\right),$$

so $F(x)$ can be obtained from $\Phi(z)$ with $z = (x - \mu)/\sigma$. Tables of $\Phi(z)$ are widely available.

Relationship Between $F(x)$ and $\Phi(z)$



Relationship between $F(x)$ and $\Phi(z)$ (source: Internet)

Skewness

The normal distribution has a symmetric shape, but many shapes are non-symmetric — a shape may be skewed to the left or to the right. The measure of skewness is

$$\texttt{skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^3 .$$

Skewness

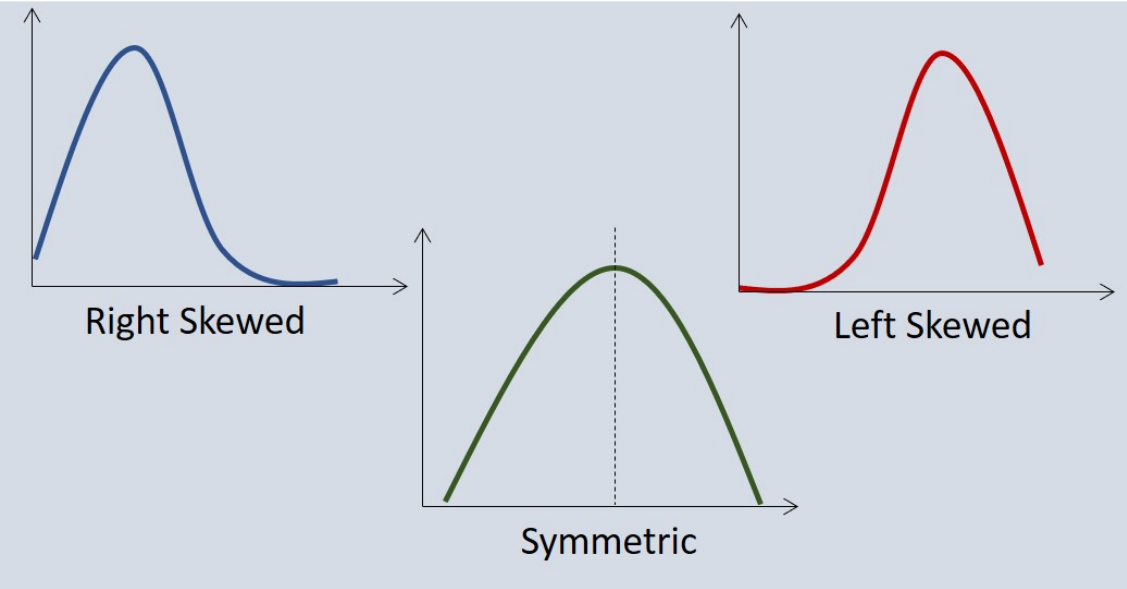
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$$\texttt{skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s} \right)^3 .$$

Skewness can be easily calculated using statistical software.

- A distribution is **skewed to the left** (negatively skewed) if its tail extends further to the left.
- A distribution is **right skewed** if its tail extends further to the right.
- A distribution is **symmetrical** if there is no obvious skewness.

Examples of Different Values of Skewness



Examples of different values of skewness.

Z-Scores

The *z-scores* give the relative location of data items within a data set. In a sample of n observations $x_1, x_2, \ldots, x_n$, the z-score z_i for observation x_i is

$$z_i = \frac{x_i - \bar{x}}{s}, \qquad i = 1, 2, \ldots, n,$$

where $\bar{x}$ is the sample mean and s the sample standard deviation. Note the similarity to the Z transformation.

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The z-score is often called the *standardized value* and is interpreted as the number of standard deviations x_i lies from the mean $\bar{x}$. For example, $z_1 = 1.7$ indicates that x_1 is 1.7 standard deviations greater than the mean; $z_1 = -0.5$ indicates it is 0.5 standard deviations smaller.

Outliers

Extreme values (unusually large or unusually small) of a data set are called *outliers*.

Identification and review of outliers is important.

Typically, any data value with a z-score less than -3 or greater than $+3$ is treated as an outlier.

Measures of Association

We are often interested in the relationship between two or more variables. Returning to the equipment store example and its scatter plot: the plot suggests a **positive relationship** between the number of commercials shown and the volume of sales, and that a straight line could approximate the relationship.

- A **positive** relationship: as the number of commercials increases, the number of sales also increases.
- A **negative** relationship: as the number of commercials increases, the number of sales decreases.

Covariance

The *covariance* is a measure of the linear relationship between two variables. For a sample of size n with observations $(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$, the sample covariance s_{xy} is

$$s_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1},$$

and the covariance of a population of size N is

$$\sigma_{xy} = \frac{\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)}{N-1}.$$

Exercise

Calculate the covariance s_{xy} for the previous example. (Answer: 11)

The Correlation Coefficient

A problem with covariance as a measure of the strength of a linear relationship is that its value depends on the units of measurement. We need a measure independent of those units.
The **correlation coefficient** r_{xy} is

$$r_{xy} = \frac{s_{xy}}{s_x s_y},$$

where s_{xy} is the sample covariance and s_x, s_y are the sample standard deviations of x and y.

Exercise

Calculate the sample correlation coefficient for the example above. Does this help with determining the strength of the relationship? (Answer: 0.93)

Population Correlation and Its Range

For population data,

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y},$$

where ρ_{xy} is the population correlation coefficient, σ_{xy} the population covariance, and σ_x, σ_y the population standard deviations.
The correlation has a limited range, $-1 \leq \rho_{xy} \leq 1$:

- $\rho_{xy} = 1$ indicates a perfect positive linear relationship;
- $\rho_{xy} = -1$ indicates a perfect negative relationship;
- $\rho_{xy} = 0$ indicates no relationship between x and y.

Exercise

Let X and Y be random variables with finite means and variances. Using the expression for ρ_{xy} above, show that $Y = a + bX$ if and only if $\rho = \pm 1$.

The Weighted Mean

The *weighted sample mean* attaches more importance (weight) to some observations than to others:

$$\bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i},$$

where x_i is the value of observation i and w_i is the weight for observation i.

Grouped Data

If data are grouped into k classes (groups or bins) with mid values $M_1, M_2, \ldots, M_k$ and corresponding class frequencies $f_1, f_2, \ldots, f_k$, the mean is

$$\bar{x} = \frac{\sum_{i=1}^{k} f_i M_i}{n},$$

and the sample variance for grouped data is

$$s^2 = \frac{\sum_{i=1}^{k} f_i (M_i - \bar{x})^2}{n-1},$$

where $f_1 + f_2 + \cdots + f_k = n$.

Summary

- Descriptive statistics organise and display raw data: frequency tables, bar charts, pie charts, histograms, ogives and scatter diagrams.
- Numerical measures describe location (mean, median, mode), variability (range, IQR, variance, standard deviation, coefficient of variation), shape (skewness) and association (covariance, correlation).
- The normal distribution and the Z transformation underpin much of what follows.
- Next: probability distributions and inferential statistics.